Chapter 11. Smoothing and Density Estimation

Histograms, kernel density estimation, bandwidth selection, multivariate KDE, confidence bands

Contents

11.1 Density estimation and histograms **2**

11.2 The kernel density estimator **2**

11.2.1 Kernels 4

11.2.2 Is the kernel density estimator a good estimator? 5

11.2.3 Choosing the bandwidth 7

11.3 Multivariate density estimation **9**

11.3.1 Multivariate density estimation with the R package `ks` 10

11.3.2 Multivariate density estimation and the curse of dimensionality 10

11.4 Confidence intervals for the density **12**

11.4.1 Pointwise confidence intervals 12

11.4.2 Confidence bands 15

Overview

This chapter and Chapter 12 discuss nonparametric function estimation by **smoothing**. So far the parameter to be estimated or tested was a single value (location, scale, a regression coefficient) or a vector (the coefficient vector in multiple regression); the parameter in these chapters is an infinite-dimensional function. The most representative cases of statistical inference where a function is the target are density estimation and nonparametric regression.

11.1 Density estimation and histograms

Given data $X_1, \ldots, X_n$ sampled from a population with density f, estimating f is **density estimation**. To estimate the distribution itself, the empirical distribution F_n estimates the distribution function F, but examining the density f, the derivative of F, is more intuitive. The empirical distribution function F_n is a step function without derivative, so it is hard to estimate the density directly from it.

Various methods have been developed for density estimation. The simplest way to visualize the empirical distribution of data as a density is the histogram, and so far we have visualized distributions by histograms. Turning the histogram into a smooth form (*smoothing* it) to get a density is the basic idea of density estimation.

Example 11.1

From data on the thickness of 485 stamps produced in 1872 in Hidalgo, a state of Mexico (`stamp.csv`), we want to estimate the number of stamp factories in Hidalgo. Under the assumption that the stamps produced by one factory have a fixed thickness, the question reduces to how many peaks (modes) the density f of the thickness of Hidalgo stamps of 1872 has.

For a rough view of the density of the Hidalgo stamp thickness data, a histogram can be used. A histogram divides the range of the data (here $(x_{\min}, x_{\max}) = (0.06, 0.131)$) into intervals and visualizes the number of observations in each, so the picture depends on the size and position of the intervals. Figure 11.1 of the book shows histograms of the same data with interval size h decreasing ($h = 0.015, 0.010, 0.005, 0.002$). How many "peaks" are visible?

The number of peaks seen varies with the interval size, so very different conclusions are possible. The human eye and brain easily ignore very small peaks (i.e. smooth naturally), so small interval sizes are better for exploring data. E.g. with the small $h = 0.002$, the very small peak between $x = 0.07$ and $x = 0.08$ is easily ignored.

Not only the interval size but also the position matters. Figure 11.2 of the book shows histograms with the interval size fixed and only the "starting point" moved slightly left; the shape changes slightly with position. The changes due to position are less marked when the intervals are small.

So if a (normalized to integrate to 1) histogram is used as a density estimator, very different estimates result depending on interval size and position. A good way to ignore the differences due to position is to average histograms, making them smooth (flat). The kernel density estimator introduced next is sometimes called a smoothed histogram.

11.2 The kernel density estimator

Kernel density estimation is the most representative method of density estimation. Before explaining it, first see how "distributions are estimated" in simpler situations:

- **Nonparametric estimation of a probability mass function.** If observations $X_1, \ldots, X_n$ follow a discrete distribution $f(x)$ ($x \in \mathcal{X} = \{0, 1, 2, \ldots, 10\}$), it suffices to examine the probability mass function at the 11 values $x \in \mathcal{X}$; for each $x \in \mathcal{X}$ the value $f(x) = P(X = x)$ can be estimated by
$$\hat{p}(x) = \hat{P}(X = x) = \frac{1}{n} \sum_{i=1}^{n} I(X_i = x). \tag{11.1}$$
This $\hat{p}(x)$ is a nonparametric estimator; since there are only finitely many parameters $\{p_x = p(x) : x \in \mathcal{X}\}$, it is very simple.
- **Nonparametric estimation of the distribution function.** Without a specific parametric model, the empirical distribution function F_n is the natural estimator of the distribution function F. It is a step function increasing by $1/n$ at each observation.

To estimate the density value $f(x)$ at a position x, there are no observations there (for continuous distributions $P(X = x) = 0$), or at most one. So we cannot count observations as in probability mass estimation. But if the density is smooth, increasing or decreasing x by small Δ_x,
$$f(x - \Delta_x) \approx f(x) \approx f(x + \Delta_x),$$
its value changes little. Kernel density estimation estimates $f(x)$ at a given x not only from observations with $X_i = x$ (which may not exist) but from all observations around x, $X_i \in (x - \Delta_x, x + \Delta_x)$. A "kernel" is a function that determines "how" the observations around a given x are used. E.g. the Gaussian kernel[1]
$$K(\Delta_x) = \frac{1}{\sqrt{2\pi}} \exp\Big(-\frac{\Delta_x^2}{2}\Big) \tag{11.2}$$
is designed so that when Δ_x is small (i.e. X_i is close to x) the kernel value is large and has high influence, and when far the kernel value is small and has negligible influence.

For a given kernel $K(\cdot)$ and a given width (*bandwidth*, analogous to the size of a histogram interval) $h > 0$, the kernel density estimator is
$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{h} K\Big(\frac{x - X_i}{h}\Big), \qquad x \in (-\infty, \infty). \tag{11.3}$$

As the density is defined for all $x \in (-\infty, \infty)$, so is its estimate, but usually $\hat{f}(x)$ is computed for x in the range of the data.

Example 11.2

To estimate the density of the population from the sodium intake data of Example 2.2 in Chapter 2, compute the kernel density estimate with the Gaussian kernel (11.2) and $h = 500$. The solid curve in Figure 11.3 of the book is the graph of $\hat{f}(x)$ (11.3) for $x \in (500, 7000)$. This density estimate is the sum of the scaled Gaussian kernels
$$\frac{1}{nh} K\Big(\frac{x - X_i}{h}\Big), \qquad i = 1, \ldots, n.$$
In Figure 11.3 the kernel functions (dashed) at $X_i = 3800, 2300$ can be found. Each kernel is highest at $x = X_i$ and decreases as x moves away from X_i. The shape of the kernel function K is chosen, and the width h decides how far away observations are used to estimate $f(x)$. E.g. the influence of $X_i = 2300$ on the estimate $\hat{f}(4000)$ is very small, judged by the very low height at $x = 4000$ of the kernel centered at that observation.

[1] The Gaussian kernel has the form of the standard normal density ("Gaussian distribution" is another name of the normal distribution). The kernel itself is not used as a distribution, so to avoid confusion the term "Gaussian kernel" is used.

Example 11.3

Kernel density estimates for the Hidalgo stamp data, again with the Gaussian kernel, comparing two widths $h = 0.001$ and 0.004. The simplest R function for the kernel density estimator is `density()`. With the data in the vector `thickness`,

```
density(thickness)
```

computes it. (The width $h \approx 0.004$ was set roughly by the "rule of thumb"; more precisely (11.12) was used; the discussion of bandwidth selection is postponed.) To make the width four times smaller than the default use

```
density(thickness, adjust = 1/4)
```

The book compares the two kernel density estimates with the histogram.

In kernel density estimation, the shape of the estimate changes greatly with the choice of the width h. Large h gives a smooth estimate; small h gives a very rough and spiky one. To explain the choice of h we must first understand the properties of the estimator determined by each value of h. Before that, kernel functions.

11.2.1 Kernels

The examples above used the Gaussian kernel, but other kernel functions can define kernel density estimates. In density estimation, a function $K : (-\infty, \infty) \to (0, \infty)$ satisfying the following is used as a kernel:

1. $K(x) \geq 0$, $\int K(x)\,dx = 1$;
2. $\lim_{|x|\to\infty} K(x) = 0$;
3. $K(x) = K(-x)$.

The first property must be satisfied by every kernel used to estimate a density; because of it the density estimator $\hat{f}(\cdot)$ has $\hat{f}(x) \geq 0$ and

$$\int \hat{f}(x)\,dx = 1,$$

so it is a density.

Proof (supplied; Exercise 11.2). $\hat{f} \geq 0$ since $K \geq 0$. Substituting $u = (x - X_i)/h$, $\int \frac{1}{h} K(\frac{x-X_i}{h}) dx = \int K(u)\,du = 1$ for each i, so $\int \hat{f} = \frac{1}{n}\sum_i 1 = 1$. ■

The second property makes the kernel large at the center (origin $x = 0$) and smaller farther from it; the third means the kernel is symmetric about the origin.[2] There are very many kinds of kernels, but the following are frequently used:

- Gaussian kernel: $K(x) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{x^2}{2})$;
- Epanechnikov kernel: $K(x) = \frac{3}{4}(1 - x^2) I(|x| \leq 1)$;
- Uniform kernel: $K(x) = \frac{1}{2} I(|x| \leq 1)$.

Figure 11.4 of the book shows these three kernels. All have the largest value at the center and smaller values away from it. The value or height of the kernel acts as a "weight": in computing

[2] A kernel need not satisfy all three properties. E.g. to estimate the "derivative" of a density, kernels taking negative values ($K(x) < 0$) can be used; depending on the form of the parameter distribution, asymmetric kernels ($K(x) \neq K(-x)$) can also be used.

$\hat{f}(x)$, observations near x get high weights and distant ones small (or 0) weights. The width h, which decides near and far, affects the estimator far more than the shape of the kernel.[3] So from now on we simply assume the Gaussian kernel.

11.2.2 Is the kernel density estimator a good estimator?

To see whether the kernel density estimator $\hat{f}(\cdot)$ (11.3) is good, we can apply the existing criteria for (numerical, not function-valued) estimators. E.g. for the estimator $\hat{\theta}_n = \bar{X}_n$ of θ with $X_i \overset{\text{i.i.d.}}{\sim} (\theta, \sigma^2)$, we looked at the bias $\text{bias}(\hat{\theta}_n) = E(\hat{\theta}_n) - \theta$ and variance $\text{Var}(\hat{\theta}_n)$ (or standard error $\text{se}(\hat{\theta}_n) = \sqrt{\text{Var}(\hat{\theta}_n)}$), and to see both at once the mean squared error $\text{MSE}(\hat{\theta}_n) = E(\hat{\theta}_n - \theta)^2$.[4] For a general estimator $\hat{\theta}_n$ the MSE splits into bias and variance:

$$\text{MSE}(\hat{\theta}_n) = E(\hat{\theta}_n - \theta)^2 = \{\text{bias}(\hat{\theta}_n)\}^2 + \text{Var}(\hat{\theta}_n). \tag{11.4}$$

Proof (supplied; Exercise 11.4). Write $\hat{\theta}_n - \theta = (\hat{\theta}_n - E\hat{\theta}_n) + (E\hat{\theta}_n - \theta)$ and square. The cross term $2(E\hat{\theta}_n - \theta)E(\hat{\theta}_n - E\hat{\theta}_n)$ vanishes, leaving $\text{Var}(\hat{\theta}_n) + \text{bias}(\hat{\theta}_n)^2$ (assuming $E\hat{\theta}_n^2 < \infty$). ■

If both bias and standard error decrease as the sample grows ($n \to \infty$), the MSE also decreases. If $\lim_{n\to\infty} \text{MSE}(\hat{\theta}_n) = 0$, the estimator $\hat{\theta}_n$ is **consistent** for θ (a consistent estimator of θ). E.g. $\hat{\theta}_n = \bar{X}_n$ has bias 0 and variance σ^2/n, so

$$\lim_{n\to\infty} \text{MSE}(\bar{X}_n) = 0^2 + \sigma^2 \lim_{n\to\infty} \frac{1}{n} = 0,$$

and $\bar{X}_n$ is a consistent estimator of $\theta = E(X_1)$.

The kernel density estimator $\hat{f}(x)$ at a given x is an estimator of the density value $f(x)$ at x. So (fixing x) the properties of $\hat{f}(x)$ as an estimator can be seen through how fast the bias $\text{bias}(\hat{f}(x))$, the variance $\text{Var}(\hat{f}(x))$ and $\text{MSE}(\hat{f}(x))$ converge to 0 as the sample grows. To see how well the "function" $\hat{f}(\cdot)$ estimates the function density $f(\cdot)$, "sum" $(\hat{f}(x) - f(x))^2$ over all x: the mean integrated squared error (MISE),

$$\text{MISE}(\hat{f}) = E\Big(\int (\hat{f}(x) - f(x))^2 \, dx \Big).$$

Bias and variance of the kernel density estimator. First, how well $\hat{f}(x)$ estimates the density value $f(x)$ at a given point x: we need the bias and variance of $\hat{f}(x)$, which are not simply expressed. So we derive approximate expressions assuming $n \to \infty$ and simultaneously $h \to 0$. Why is the width h assumed to shrink as the sample grows (as in large-sample approximation)? The kernel density estimator $\hat{f}(\cdot)$ (11.3) depends greatly on the choice of h. With large h, many observations are "included" in computing $\hat{f}(x)$ at a given x; with small h, few. If the sample size n grows, the number of observations included in a kernel of fixed width grows too, so h is designed to shrink as n grows.

First assume the population density $f(x)$ is a smooth function:

> **Assumption 1.** The second derivative f'' of the density is absolutely continuous, and the third derivative satisfies $\int (f'''(x))^2 \, dx < \infty$.

[3] When the sample is large enough, the effects of the width h and of the kind of kernel can be compared through Theorems 11.1, 11.2 and 11.3.

[4] The mean squared error is the expectation (risk) of the loss function $L(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$: $\text{MSE}(\hat{\theta}) = R(\hat{\theta}, \theta) = E\{L(\hat{\theta}, \theta)\}$. Papers and books often say "risk" instead of mean squared error.

Under Assumption 1, near a point x the density f can be approximated by a quadratic function. By Taylor expansion, for t close to x,

$$f(t) = f(x) + f'(x)(t-x) + \frac{f''(x)}{2}(t-x)^2 + R(t-x),$$

where the last term $R(t-x) = o((t-x)^2)$, i.e. $\lim_{t\to x} R(t-x)/(t-x)^2 = 0$.[5] Writing t near x as $t = x - hu$,

$$f(x-hu) = f(x) - hf'(x)u + \frac{h^2 f''(x)}{2}u^2 + o((hu)^2). \tag{11.5}$$

Theorem 11.1

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} f$, with the population density f satisfying Assumption 1. At a point x with $f(x) > 0$, the bias and variance of the kernel density estimator $\hat{f}(x)$, as $n \to \infty$, $h \to 0$, are

$$\operatorname{bias}(\hat{f}(x)) = \frac{f''(x)\sigma_K^2}{2}h^2 + o(h^2), \tag{11.6}$$

$$\operatorname{Var}(\hat{f}(x)) = \frac{1}{nh}f(x)\int K^2(u)\,du + o\Big(\frac{1}{nh}\Big). \tag{11.7}$$

The bias is large when f is very complicated, i.e. when $|f''(x)|$ is large. To reduce the bias one can make the width h small, but then the variance becomes large. Conversely, enlarging h to make the estimator flatter reduces the variance but increases the bias. This phenomenon, the **bias–variance trade-off**, occurs not only in density estimation but commonly in many estimation problems. Remember that, as in (11.4), the squared bias plus the variance is the MSE; Figure 11.5 of the book illustrates the trade-off.

Proof of Theorem 11.1. First the bias:

$$\operatorname{bias}(\hat{f}(x)) = E(\hat{f}(x)) - f(x) = \frac{1}{n}\sum_{i=1}^{n} E\Big[\frac{1}{h}K\Big(\frac{x-X_i}{h}\Big)\Big] - f(x). \tag{11.8}$$

Regarding the kernel as a density, its first and second moments (mean and variance) can be computed. By the third property of kernels, "symmetry,"

$$\mu_K = \int xK(x)\,dx = 0, \qquad \sigma_K^2 = \int (x-0)^2 K(x)\,dx.$$

Using this and the approximation (11.5) of the density,

$$\begin{aligned} E\Big[\frac{1}{h}K\Big(\frac{x-X_i}{h}\Big)\Big] &= \int \frac{1}{h}K\Big(\frac{x-t}{h}\Big)f(t)\,dt = \int K(u)f(x-hu)\,du \\ &= \int K(u)\Big[f(x) - hf'(x)u + \frac{h^2 f''(x)}{2}u^2 + o((hu)^2)\Big]du \\ &= f(x) + \frac{f''(x)\sigma_K^2}{2}h^2 + o(h^2), \end{aligned}$$

and substituting into (11.8) gives (11.6).[6]

[5] A sequence a_n is $a_n = o(n^{-1})$ ("little o") if $\lim_{n\to\infty} a_n/n^{-1} = 0$, i.e. a_n converges to 0 faster than n^{-1}. More generally, functions g, h satisfy $g(x) = o(h(x))$ as $x \to \infty$ if $\lim_{x\to\infty} g(x)/h(x) = 0$.

[6] Integrating "little o" or "big O" does not always give the same term again: $\int o(h^2u^2)\,du = o(h^2)$ is not always true. To be exact, the error of the second-order approximation of $f(x)$ must be expressed exactly with Taylor's theorem, and the order of lim and $\int$ exchanged using the dominated convergence theorem or similar. (Supplied: under Assumption 1, f'' is bounded and uniformly continuous on compact sets, and with $\int u^2K(u)\,du < \infty$ the remainder $\int K(u)[f(x-hu) - f(x) + huf'(x) - \frac{h^2u^2}{2}f''(x)]du$ is $h^2\int K(u)u^2\,\epsilon(hu)\,du$ with $\epsilon(s) \to 0$ and bounded; dominated convergence gives $o(h^2)$.)

The variance is computed similarly. For simplicity use "big O" notation for $n \to \infty$, $h \to 0$; e.g. $h^2 + o(h^2) = O(h^2)$, $\frac{1+h+h^2}{n} = O(\frac{1}{n})$.

$$\begin{aligned}\operatorname{Var}(\hat{f}(x)) &= \frac{1}{n}\operatorname{Var}\Big[\frac{1}{h}K\Big(\frac{x-X_i}{h}\Big)\Big] \\ &= \frac{1}{n}\Big[E\Big\{\frac{1}{h}K\Big(\frac{x-X_i}{h}\Big)\Big\}^2 - f^2(x) + O(h^2)\Big] \\ &= \frac{1}{nh}f(x)\int K^2(u)\,du + O\Big(\frac{1}{n}\Big).\end{aligned}$$

(Supplied detail: $E\{\frac{1}{h^2}K^2(\frac{x-X_i}{h})\} = \frac{1}{h}\int K^2(u)f(x-hu)\,du = \frac{1}{h}f(x)\int K^2 + O(1)$, and the squared mean is $f(x)^2 + O(h^2)$.) Since $h \to 0$, $O(\frac{1}{n})$ is $o(\frac{1}{nh})$, which gives (11.7). ■

To reduce the MSE both bias and variance must shrink: as the sample grows ($n \to \infty$), the width must shrink ($h \to 0$), but not faster than the sample grows ($nh \to \infty$).

Theorem 11.2

When $n \to \infty$, $h \to 0$, $nh \to \infty$, the mean squared error and mean integrated squared error of the kernel density estimator $\hat{f}(x)$ are

$$\begin{aligned}\operatorname{MSE}(\hat{f}(x)) &= \frac{(f''(x)\sigma_K^2)^2}{4}h^4 + \frac{f(x)}{nh}\int K^2(u)\,du + o(h^4) + o\Big(\frac{1}{nh}\Big), \\ \operatorname{MISE}(\hat{f}) &= \frac{\sigma_K^4\int(f''(x))^2\,dx}{4}h^4 + \frac{1}{nh}\int K^2(u)\,du + o(h^4) + o\Big(\frac{1}{nh}\Big).\end{aligned} \tag{11.9}$$

Proof (supplied). The first line is (11.4) with Theorem 11.1. For the second, $\operatorname{MISE}(\hat{f}) = \int \operatorname{MSE}(\hat{f}(x))\,dx$ by Fubini; integrating the leading terms uses $\int f = 1$, and the remainders integrate to $o(h^4) + o(\frac{1}{nh})$ under Assumption 1 (by the same dominated-convergence argument applied uniformly in x). ■

By Theorem 11.2, when all three conditions $n \to \infty$, $h \to 0$, $nh \to \infty$ hold, $\operatorname{MSE}(\hat{f}(x)) \to 0$, so $\hat{f}(x)$ is a consistent estimator of $f(x)$; also $\operatorname{MISE}(\hat{f}) \to 0$, so the kernel density estimator $\hat{f}$ is a consistent estimator of the density f.

11.2.3 Choosing the bandwidth

In the expression (11.9) of the MISE, given the kernel K, the density f to be estimated and the sample size n, the value of h minimizing $\operatorname{MISE}(\hat{f})$ can be found. Ignoring the two asymptotically negligible terms $o(h^4) + o(\frac{1}{nh})$ and regarding $\operatorname{MISE}(\hat{f})$ as a function of h, the point where its derivative is 0 (the value of h where the MSE in Figure 11.5 is smallest) is

$$\frac{d}{dh}\operatorname{MISE}(\hat{f}) = 0 \iff h^* = \Big(\frac{\int K^2(u)\,du}{\sigma_K^4\int(f''(x))^2\,dx}\cdot\frac{1}{n}\Big)^{\frac{1}{5}}.$$

(Supplied: $\frac{d}{dh}[\frac{A}{4}h^4 + \frac{B}{nh}] = Ah^3 - \frac{B}{nh^2} = 0$ gives $h^5 = \frac{B}{An}$, with $A = \sigma_K^4\int(f'')^2$ and $B = \int K^2$; the second derivative is positive, so this is the minimum.) This h^* is the asymptotically optimal bandwidth. This theoretical width depends on the kernel K and the second derivative f'' of the population density, but as the sample size n grows it shrinks at roughly the rate $h^* \approx n^{-1/5}$.[7] If K is the Gaussian kernel,

$$\int K^2(u)\,du = \frac{1}{2\sqrt{\pi}}, \qquad \sigma_K^2 = 1,$$

so

$$h^* = \Big(\frac{1}{2\sqrt{\pi}\int(f''(x))^2\,dx}\cdot\frac{1}{n}\Big)^{\frac{1}{5}}, \tag{11.10}$$

[7] At $n = 100$, $n^{-1/5} = 0.39$; at $n = 10000$, $n^{-1/5} = 0.16$: the width shrinks much more slowly than the sample size grows.

but this value can be known only if the unknown parameter f is known. Various methods have been proposed depending on how $\int (f''(x))^2\,dx$ is estimated; three representative ones are introduced here.

Normal reference rule. If $f(x) = \frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$, i.e. the population is normal $N(\mu, \sigma)$, it is known that

$$\int (f''(x))^2\,dx = \frac{3}{8\sqrt{\pi}\sigma^5}. \tag{11.11}$$

Proof of (11.11) (supplied). For $\sigma = 1$, $\phi''(x) = (x^2-1)\phi(x)$ and $\phi(x)^2 = \frac{1}{2\sqrt{\pi}} \cdot \frac{1}{\sqrt{2\pi}\cdot 2^{-1/2}} e^{-x^2}$ is $\frac{1}{2\sqrt{\pi}}$ times the $N(0, \frac{1}{2})$ density. So $\int(\phi'')^2 = \frac{1}{2\sqrt{\pi}} E(Z^2-1)^2$ with $Z \sim N(0, \frac{1}{2})$: $EZ^4 - 2EZ^2 + 1 = \frac{3}{4} - 1 + 1 = \frac{3}{4}$, giving $\frac{3}{8\sqrt{\pi}}$. For general σ, $f''(x) = \sigma^{-3}\phi''((x-\mu)/\sigma)$, so $\int(f'')^2 = \sigma^{-5}\int(\phi'')^2$. ■

Substituting (11.11) into (11.10) gives $h^* = (4/3)^{1/5} n^{-1/5}\sigma$, and replacing the population variance σ^2 by its estimate gives Scott's (Scott, 2015) normal reference rule (nrd):

$$h_n^{\text{nrd}} = 1.06 n^{-\frac{1}{5}}\hat{\sigma}, \qquad \hat{\sigma} = \min\Big\{s, \frac{\text{IQR}}{1.34}\Big\},$$

where the estimate $\hat{\sigma}$ of the population standard deviation is the smaller of the sample standard deviation s and the robust estimator IQR/1.34. The IQR is the distance between the 75% quantile $Q_n(3/4)$ and 25% quantile $Q_n(1/4)$ of the data, IQR $= Q_n(3/4) - Q_n(1/4)$. If the population is $N(\mu, \sigma)$, IQR/1.34 is a consistent estimator of σ. (Supplied: the IQR of $N(\mu, \sigma^2)$ is $2 \cdot 0.6745\sigma = 1.349\sigma$.)

"Rule of thumb." For populations not following a normal distribution, h_n^{nrd} from the normal reference rule tends to be a little too large and is generally inefficient, so the width is adjusted smaller:

$$h_n^{\text{nrd0}} = 0.9 n^{-\frac{1}{5}}\hat{\sigma}, \qquad \hat{\sigma} = \min\Big\{s, \frac{\text{IQR}}{1.34}\Big\}. \tag{11.12}$$

This is Silverman's (Silverman, 1986) rule-of-thumb method. Silverman's rule of thumb is fast and easy to compute and the most used, but it also assumes a density f similar to the normal.

Plug-in estimation. This method does not assume the form of the density f in advance but directly estimates the unknown parameter $A(f) \equiv \int (f''(x))^2\,dx$. Sheather and Jones (1991) proposed estimating $A(f)$ directly using another kernel density estimate and plugging it in. With this estimate $\hat{A}(f)$,[8] the Sheather–Jones plug-in estimator is

$$h_n^{\text{SJ}} = \Big(\frac{1}{2\sqrt{\pi}\hat{A}(f)} \cdot \frac{1}{n}\Big)^{\frac{1}{5}}.$$

Besides estimating and plugging in the unknown $A(f)$, there are various other bandwidth selection methods, e.g. using cross-validation. Silverman's rule of thumb is easy to understand and fast, so in most cases it is the first choice. Generally the Sheather–Jones plug-in estimate is recommended as the width h of kernel density estimation. In R, the normal reference width h_n^{nrd} is computed by `bw.nrd()`, the rule-of-thumb width h_n^{nrd0} by `bw.nrd0()`, and the plug-in width h_n^{SJ} by `bw.SJ()`. The width can also be specified inside `density()`, which computes density estimates. The next example compares the three methods.

[8] The Sheather–Jones estimate $\hat{A}(f)$ is expressed by an integral of another kernel estimate of the rth derivative $f^{(r)}$ of f ($r = 1, 2, 3, 4$). Estimating derivatives of a density is harder than estimating the density, and those estimates are more variable.

Example 11.4

Kernel density estimates for the Hidalgo stamp data with the three widths above. As in the code below, specify the argument `bw = "nrd0"`; the results are in Figure 11.6 of the book. The density of these data has several peaks and differs greatly from a normal density, so the Sheather–Jones plug-in width $h_n^{\text{SJ}} = 0.0012$ seems the most appropriate.

```
> bw.SJ(thickness)
[1] 0.001205108
> kde.nrd = density(thickness, bw = "nrd")
> kde.nrd0 = density(thickness, bw = "nrd0") # default bw = "
    nrd0"
> kde.SJ = density(thickness, bw = "SJ")
```

11.3 Multivariate density estimation

A multivariate or joint density f is a function expressing the joint distribution of p continuous random variables $X_1, \ldots, X_p$ in the form

$$P((X_1, \ldots, X_p) \in A) = \int_A f(x_1, \ldots, x_p)\, dx_1 \cdots dx_p.$$

Given vector observations $\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{i.i.d.}}{\sim} f$ (here $\mathbf{X}_i = (X_{i1}, \ldots, X_{ip})'$), we want to estimate the joint density f from them. The multivariate kernel density estimator is defined by directly extending the univariate one.

As in one dimension, to estimate $f(\mathbf{x})$ at a given $\mathbf{x} = (x_1, \ldots, x_p)'$ we use observations around $\mathbf{x}$. This neighborhood is part of p-dimensional space, so the univariate kernel is extended to a multivariate kernel. The most representative Gaussian kernel has the form of the multivariate standard normal density:

$$K(\mathbf{x}) = (2\pi)^{-\frac{p}{2}} \exp\big(-\tfrac{1}{2}\mathbf{x}'\mathbf{x}\big).$$

The multivariate kernel is highest at the origin $\mathbf{0}$ and symmetric about it, and its size can be changed by a bandwidth matrix H. E.g. from the multivariate Gaussian kernel K, defining

$$K_H(\mathbf{x}) := |H|^{-1/2} K(H^{-1/2}\mathbf{x})$$

gives the joint density of the normal $N_p(\mathbf{0}, H)$. To see the effect of the bandwidth matrix H (with the Gaussian kernel, the covariance matrix of the normal) in a multivariate kernel, Figure 11.7 of the book shows the Gaussian kernel for $p = 2$ with different H:

$$H_a = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad H_b = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad H_c = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad H_d = \begin{pmatrix} 1 & -.8 \\ -.8 & 1 \end{pmatrix}.$$

E.g. with the kernel bandwidth matrix H_b, more smoothing is done in the direction of the first variable x_1 and less in the direction of x_2.

Besides the Gaussian kernel, other kernels can be used; e.g. the uniform kernel generalizes to the box-shaped kernel

$$K^{\text{Unif}}(\mathbf{x}) = K^{\text{Unif}}(x_1, \ldots, x_p) = \tfrac{1}{2} I(|x_1| \le 1) \times \cdots \times \tfrac{1}{2} I(|x_p| \le 1).$$

Given the kernel K and bandwidth matrix H, the multivariate kernel density estimator from the sample $\mathbf{X}_1, \ldots, \mathbf{X}_n$ is

$$\hat{f}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^{n} K_H(\mathbf{x} - \mathbf{X}_i). \tag{11.13}$$

11.3.1 Multivariate density estimation with the R package `ks`

To do multivariate density estimation with the kernel estimator (11.13), the kernel bandwidth matrix H must be set. This section shows an example of kernel density estimation with the R package `ks`, which computes the plug-in estimate of the optimal bandwidth matrix[9] and provides several auxiliary visualization tools. (The same `ks` package can of course also estimate univariate densities.)

Example 11.5

Using data on the size of several penguin species in the Palmer Archipelago of Antarctica (`penguins.csv`), estimate the joint density of two variables: bill length and flipper length. The scatter plot of the two variables, colored by the three species surveyed (Adelie, Chinstrap, Gentoo), is in Figure 11.8 of the book.

Ignoring species information, we estimate the density of bill and flipper length for all penguins. First, `ks::Hpi()` gives the plug-in estimate $\hat{H}$ of the bandwidth matrix:

```
> penguins = read.csv("data/penguins.csv")
> penguins.b = penguins %>%
+   select(bill_length_mm, flipper_length_mm)
> library(ks)
> H <- ks::Hpi(penguins.b)
> H
         [,1]      [,2]
[1,] 2.618165  3.454826
[2,] 3.454826 13.736760
```

The bandwidth matrix $\hat{H}$ roughly matches the dispersion (covariance matrix) of all the data. The kernel density estimate (11.13) with this bandwidth matrix is computed by `ks::kde()`; its result can be visualized with the `plot()` function redefined in the `ks` package. Two examples of visualization are in Figure 11.9 of the book.

```
kde.p <- ks::kde(x = penguins.b, H = H)
# contour of density estimate.
plot(kde.p, display = "filled.contour", cont = seq(5, 95, by = 10),
     xlab = "bill length", ylab = "flipper length")
points(kde.p$x, pch = 1, cex = 0.5)
# 3D plot
plot(kde.p, display = "persp", xlab = "bill_length", ylab = "flipper
    length")
```

11.3.2 Multivariate density estimation and the curse of dimensionality

Smoothing is a very powerful way to estimate parameters (here densities) in parameter spaces of uncountably high dimension, and the key to its success is that as the kernel width $h \to 0$, the number of observations included in the kernel K, $nh \to \infty$, does not decrease but grows. The problem that commonly arises when applying smoothing to multivariate data is precisely the **curse of dimensionality**.

Why it arises, by a simple example. First suppose univariate observations from Uniform$(0, 1)$. With the uniform kernel of width 0.05, the number of observations in the kernel is on average $0.1 \times n = n/10$. If the dimension increases by one and bivariate observations come from the

[9] This book does not derive the optimal bandwidth matrix in multivariate density estimation. The bandwidth matrix H^* minimizing asymptotically the MSE of Theorem 11.3 is the optimal one, and as in Section 11.2.3 the plug-in estimator can be derived by similar logic (Wand and Jones, 1994).

uniform distribution on $(0,1)\times(0,1)$, the average number of observations in the uniform kernel with bandwidth matrix $H=(0.05)^2 I$ is $(0.1)^2 n = n/100$. If 10-dimensional observations come from the uniform distribution on $(0,1)^p$ $(p=10)$, the average number of observations in the uniform kernel with $H=(0.05)^2 I$ is

$$(0.1)^p n = \frac{n}{10000000000},$$

so ten billion observations are needed for the computation of a 10-dimensional kernel density estimate $\hat{f}(\mathbf{x})$ (11.13) to include on average one observation. Nonparametric estimation of multivariate data needs a very large number of observations. This is the statistical curse of dimensionality.[10]

Consider the influence of the curse of dimensionality on multivariate density estimation. Suppose the ith diagonal element of the bandwidth matrix H is h_i^2 and all other elements are 0 (this restriction is often used in practice):

$$H = \begin{pmatrix} h_1^2 & 0 & \cdots & 0 \\ 0 & h_2^2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & h_p^2 \end{pmatrix},$$

and

$$K_H(\mathbf{x}) = \frac{1}{h_1} K\Big(\frac{x_1}{h_1}\Big)\cdots\frac{1}{h_p} K\Big(\frac{x_p}{h_p}\Big), \tag{11.14}$$

where $K(x)$ on the right is the kernel defined in one dimension, or the multivariate kernel with $p=1$. The influence of the dimension p on the MSE of the multivariate kernel density estimator at each point $\mathbf{x}$:

Theorem 11.3 (García-Portugués (2021), Theorem 3.2)

With a kernel satisfying (11.14), the MSE of $\hat{f}(\mathbf{x})$ is

$$\mathrm{MSE}(\hat{f}(\mathbf{x})) = \Big[\frac{\sigma_K^2}{2}\sum_{j=1}^{p} h_j^2 A_j(f)\Big]^2 + \frac{(\int K^2(u)\,du)^p}{nh_1\cdots h_p} + o\Big(\sum_j h_j^4\Big) + o\Big(\frac{1}{nh_1\cdots h_p}\Big), \tag{11.15}$$

where $A_j(f) = \frac{\partial^2}{\partial x_j \partial x_j} f(\mathbf{x})$ (the book writes $\int(\frac{\partial^2}{\partial x_j \partial x_j} f(\mathbf{x}))^2 d\mathbf{x}$, which belongs to the integrated version), and $K(u)$ is the one-dimensional kernel of (11.14).

Sketch (supplied). As in Theorem 11.1, $E\hat{f}(\mathbf{x}) = \int \prod_j K(u_j) f(\mathbf{x} - (h_1u_1,\dots,h_pu_p)')\,d\mathbf{u}$. A second-order Taylor expansion in p variables, with $\int u_j K(u_j) = 0$, $\int u_j^2 K(u_j) = \sigma_K^2$ and the cross terms $u_j u_k$ $(j\neq k)$ integrating to 0 by independence of coordinates under the product kernel, gives bias $\frac{\sigma_K^2}{2}\sum_j h_j^2 \partial_{jj} f(\mathbf{x}) + o(\sum_j h_j^2)$. The variance is $\frac{1}{n}\operatorname{Var} K_H(\mathbf{x}-\mathbf{X}_1) = \frac{f(\mathbf{x})}{nh_1\cdots h_p}(\int K^2)^p + O(\frac{1}{n})$, since $\int K_H^2 = (h_1\cdots h_p)^{-1}(\int K^2)^p$. (The variance term carries the factor $f(\mathbf{x})$, omitted in the book's display.) Squaring the bias and adding the variance gives (11.15). ∎

For simplicity take all widths equal ($h_1=\cdots=h_p=h$). With the dimension p fixed, the optimal width minimizing $\mathrm{MSE}(\hat{f}(\mathbf{x}))$ is $h^* = O(n^{-1/(4+p)})$. Substituting it back into (11.15),

$$\mathrm{MSE}(\hat{f}(\mathbf{x})) = O(n^{-\frac{4}{4+p}}),$$

so when the dimension p is high the MSE decreases to 0 very slowly. (Supplied: balancing h^4 against $\frac{1}{nh^p}$ gives $h^{4+p} \asymp \frac{1}{n}$, and then both terms are $\asymp n^{-4/(4+p)}$.) That is, when the dimension is high, the sample size needed to ensure the accuracy of multivariate density estimation is very large. Density estimation for one- or two-dimensional observations is often used, but for three or more dimensions its practicality drops greatly.

[10] The curse of dimensionality appears in various forms in multivariate statistical analysis. Wasserman (2006) called "the statistical curse of dimensionality" situations where the required sample size grows geometrically with the dimension, or where the sample looks large but the dimension is so high that estimates are very uncertain.

11.4 Confidence intervals for the density

So far we have looked at density estimation and its asymptotic properties; in particular, with suitable choice of kernel and width, density estimation is asymptotically accurate. Then, when the sample is not infinitely large, what statistical inference is possible about the uncertainty of the estimated density? This question can be answered by giving confidence intervals for the estimated density. Here we look at intervals for the univariate kernel density estimator.

The density f has a value $f(x)$ at each point x. A $1-\alpha$ confidence interval for the density value $f(x)$ is an interval $C_{1-\alpha}(x)$ computable from the data such that the probability that the actual density value $f(x)$ lies in $C_{1-\alpha}(x)$ is at least $1-\alpha$. The parameter f is a function with values at infinitely many x, so two definitions are used:

- **Pointwise confidence interval:** for a given position x, the interval $C_{1-\alpha}(x)$ satisfies
$$P(f(x) \in C_{1-\alpha}(x)) \geq 1-\alpha. \tag{11.16}$$
- **Confidence band:** the intervals $C_{1-\alpha}(x)$ satisfy
$$P(f(x) \in C_{1-\alpha}(x),\ \forall x) \geq 1-\alpha. \tag{11.17}$$

The pointwise interval guarantees coverage locally at a given position x; the band guarantees coverage globally, for all x or the whole function $f(\cdot)$. In nonparametric models with few restrictions on the form of f an exact interval for the density is nearly impossible, so asymptotically exact intervals are sought. Also, every kernel density estimator $\hat{f}(x)$ has bias. In univariate density estimation, e.g., even with the optimal width $h^* = O(n^{-1/5})$ the bias is $O(n^{-2/5})$, and the standard error of the estimator is also $O(n^{-2/5})$, a non-negligible size. Methods for bias-adjusted (removed) intervals have been developed,[11] but they exceed the level of this book; here we estimate not the original density $f(x)$ but the smoothly mollified density

$$f_h(x) = E(\hat{f}(x)),$$

where h is the width of the kernel density estimator. This width is generally estimated from the data, but once set it is treated as given.

Estimating the mollified density is actually much used, especially effective when the density f has a very wiggly shape, for grasping the shape and topological features of the distribution (Figure 11.10 of the book).

11.4.1 Pointwise confidence intervals

The kernel density estimator $\hat{f}$ has the form of a mean of about $2nh$ observations included in the kernel (exactly a mean with the uniform kernel, a weighted mean with the Gaussian kernel). If $nh \to \infty$, by the CLT

$$\sqrt{nh}(\hat{f}(x) - f_h(x)) \dot{\sim} N(0, \sigma_f^2(x)), \tag{11.18}$$

where $\sigma_f^2(x) = f(x)a_K$, $a_K = \int K^2(u)\,du$. With the Gaussian kernel $a_K = 1/(2\sqrt{\pi})$, with the uniform kernel $a_K = 1/4$. The large-sample interval for the mollified density value $f_h(x)$ at x is obtained from (11.18); the unknown $\sigma_f^2(x)$ or $f(x)$ is replaced by its estimate.

Plug-in large-sample interval. Substituting $\hat{f}(x)$ for $f(x)$,

$$f_h(x) \in \hat{f}(x) \pm z_{\alpha/2}\sqrt{\frac{\hat{f}(x)a_K}{nh}}$$

is an asymptotically exact pointwise interval.

[11] Bias-removed intervals are most easily introduced in Chen (2017).

Example 11.6

For the Hidalgo stamp data, using the kernel density estimate $\hat{f}(x)$ with width $h = h_n^{\mathrm{SJ}} = 0.0012$, compute plug-in large-sample pointwise intervals at each point x of $f(x)$. The code below uses `ks::kde()` for the estimate `ks.out$estimate`, then the plug-in large-sample intervals. `ks.out$estimate` is the vector of $\hat{f}(x_i)$ at the points $x_1, \ldots, x_M$ defined in `ks.out$eval.points`, so `ci.lower` and `ci.upper` below are vectors of the same length.

```
> h = bw.SJ(thickness)
> n = length(thickness)
> ks.out = ks::kde(thickness, h = h)
> f.hat = ks.out$estimate
> ci.lower = f.hat - qnorm(1-0.025) * sqrt( f.hat / (2*sqrt(pi))
    / (n*h))
> ci.upper = f.hat + qnorm(1-0.025) * sqrt( f.hat / (2*sqrt(pi))
    / (n*h))
> data.frame(x = ks.out$eval.points, ci.low = ci.lower, ci.up =
    ci.upper) %>% head()
          x      ci.low      ci.up
1 0.05554110 -0.03660399 0.03810936
2 0.05574090 -0.04897191 0.05170534
3 0.05594069 -0.06449555 0.06932493
4 0.05614048 -0.08357777 0.09187994
5 0.05634028 -0.10651872 0.12040574
6 0.05654007 -0.13344939 0.15605133
```

This kernel density estimate and pointwise 95% intervals are shown in Figure 11.11 of the book.

Bootstrap plug-in interval. Estimate $\sigma_f^2(x)$ by the bootstrap. Rewriting (11.18),

$$\hat{f}(x) \dot{\sim} N(f_h(x), \mathrm{se}^2(\hat{f}(x))),$$

so compute the nonparametric bootstrap estimate of $\mathrm{se}(\hat{f}(x))$. Given a grid $\mathcal{X} = \{x_1, \ldots, x_M\}$ finely dividing the interval containing the observations, the statistics $\hat{f}(x)$ at $x \in \mathcal{X}$ are all estimates from one sample $X_1, \ldots, X_n$, so bootstrap resampling can be done once for all $x_i \in \mathcal{X}$. Generate for each x_i the bootstrap replicates

$$\hat{f}^{*1}(x_i), \ldots, \hat{f}^{*B}(x_i), \tag{11.19}$$

take their sample standard deviation as the bootstrap estimate of the standard error,

$$\widehat{\mathrm{se}}^{\mathrm{Boot}}(\hat{f}(x)) = \Big(\frac{\sum_{b=1}^{B} (\hat{f}^{*b}(x_i) - \overline{\hat{f}^{*}(x_i)})^2}{B-1} \Big)^{1/2},$$

and the $1-\alpha$ bootstrap plug-in interval is

$$f_h(x) \in \hat{f}(x) \pm z_{\alpha/2} \widehat{\mathrm{se}}^{\mathrm{Boot}}(\hat{f}(x)).$$

Example 11.7

With the data of Example 11.6, the bootstrap plug-in interval:

```
x = ks.out$eval.points
kde0 = ks.out$estimate
n = length(thickness)

B = 1999
```

```
kde.boot = replicate(B,{
          ks::kde(sample(thickness,n, replace = TRUE),
          h = h,           # same bandwidth h
          eval.points = x # same points x
          )$estimate       # compute kde(x)
})
se.fhat = apply(kde.boot,1,sd)
ci.lower = f.hat - qnorm(1-0.025) * se.fhat
ci.upper = f.hat + qnorm(1-0.025) * se.fhat
```

The result is shown in Figure 11.11 of the book.

Most of the bootstrap intervals of Chapter 8 apply, but the number of points to examine ($M = \#\mathcal{X}$) is large, so the computation grows. The simplest, the basic bootstrap interval, is often used.

Basic bootstrap interval. From the bootstrap replicates (11.19) at each point x, compute the $1-\frac{\alpha}{2}$ and $\frac{\alpha}{2}$ quantiles $Q_{\hat{f}^*}(1-\frac{\alpha}{2};x)$, $Q_{\hat{f}^*}(\frac{\alpha}{2};x)$; the $1-\alpha$ basic bootstrap interval is

$$f_h(x) \in \big(2\hat{f}(x) - Q_{\hat{f}^*}(1-\tfrac{\alpha}{2};x),\ 2\hat{f}(x) - Q_{\hat{f}^*}(\tfrac{\alpha}{2};x)\big).$$

It approximates the distribution of $\hat{f}(x) - f_h(x)$ by the bootstrap $\hat{f}^*(x) - \hat{f}(x)$.

Absolute-deviation bootstrap interval. If the distribution of the deviation $\hat{f}(x) - f_h(x)$ can be assumed symmetric (asymptotically true), the distribution of the absolute deviation $|\hat{f}(x) - f_h(x)|$ can be bootstrap-approximated. The bootstrap replicates are

$$\Delta^{*1}(x) = |\hat{f}^{*1}(x) - \hat{f}(x)|,\ \ldots,\ \Delta^{*B}(x) = |\hat{f}^{*B}(x) - \hat{f}(x)|,$$

and with their $1-\alpha$ quantile $Q_{\Delta^*}(1-\alpha;x)$, the $1-\alpha$ absolute-deviation bootstrap interval is

$$f_h(x) \in \big(\hat{f}(x) - Q_{\Delta^*}(1-\alpha;x),\ \hat{f}(x) + Q_{\Delta^*}(1-\alpha;x)\big).$$

(Note the $1-\alpha$ quantile is used, not $1-\frac{\alpha}{2}$.) All intervals introduced so far are asymptotically exact; e.g. the absolute-deviation bootstrap interval, as $n \to \infty$, $h \to 0$, $nh \to \infty$, $B \to \infty$, satisfies

$$P\big[f_h(x) \in \big(\hat{f}(x) - Q_{\Delta^*}(1-\alpha;x),\ \hat{f}(x) + Q_{\Delta^*}(1-\alpha;x)\big)\big] = 1 - \alpha + o(1).$$

Why the $1-\alpha$ quantile (supplied; Exercise 11.6). If $|\hat{f}(x) - f_h(x)| \sim G$ with G continuous and invertible, then $P(|\hat{f}(x) - f_h(x)| \le G^{-1}(1-\alpha)) = 1-\alpha$, i.e. $P(f_h(x) \in \hat{f}(x) \pm G^{-1}(1-\alpha)) = 1-\alpha$. The absolute value already folds both tails, so the $1-\alpha$ (not $1-\frac{\alpha}{2}$) quantile is used. If the bootstrap distribution G^* of $|\hat{f}^*(x) - \hat{f}(x)|$ equals G, the bootstrap interval is exact; asymptotically $G^* \to G$. ■

Example 11.8

Basic bootstrap intervals from the replicates `kde.boot` of Example 11.7:

```
kde.quantiles = apply(kde.boot, 1,
                      function(x){quantile(x,probs = c
                      (1-0.025,0.025))})
ci.lower = 2*f.hat - kde.quantiles[1,]
ci.upper = 2*f.hat - kde.quantiles[2,]
```

For the absolute-deviation bootstrap interval, first compute the bootstrap replicates of the absolute deviations at all $x \in \mathcal{X}$:

```
abs.dev.boot = apply(kde.boot, 2, function(x){ abs(x - f.hat) })
abs.dev.boot.quantiles = apply(abs.dev.boot, 1,
                         function(x){quantile(x,probs =
                         1-0.05)})
ci.lower = f.hat - abs.dev.boot.quantiles
ci.upper = f.hat + abs.dev.boot.quantiles
```

Figure 11.11 of the book shows the four pointwise intervals computed so far. The plug-in large-sample interval is a little longer, but they are broadly similar.

The intervals of Figure 11.11 are (asymptotically) exact at each point x, but the graph of the density f cannot be asserted to lie inside the intervals at the same level: (11.16) is satisfied but (11.17) may not be. Bands satisfying (11.17) are treated next.

Remark

To simplify the standard error estimate so that no additional computation is needed,

$$\frac{\hat f(x) - f_h(x)}{\widehat{\mathrm{se}}(\hat f(x))}, \qquad \widehat{\mathrm{se}}(\hat f(x)) = \hat\sigma_f(x)/\sqrt{nh} = \sqrt{\frac{\hat f(x) a_K}{nh}}, \tag{11.20}$$

is asymptotically a pivot, so the standard bootstrap-t interval can be used. The distribution of the pivot (11.20) is bootstrap-approximated by

$$t^*(x) \equiv \frac{\hat f^*(x) - \hat f(x)}{\sqrt{\frac{\hat f^*(x) a_K}{nh}}} \mathrel{\dot\sim} \frac{\hat f(x) - f_h(x)}{\widehat{\mathrm{se}}(\hat f(x))}.$$

With $Q_{t^*}(p; x)$ the p-quantile of the bootstrap replicates $t^*(x)$, the bootstrap-t interval is

$$f_h(x) \in \Big(\hat f(x) - Q_{t^*}(1-\tfrac{\alpha}{2}; x)\sqrt{\frac{\hat f(x) a_K}{nh}},\ \hat f(x) - Q_{t^*}(\tfrac{\alpha}{2}; x)\sqrt{\frac{\hat f(x) a_K}{nh}}\Big).$$

However, if $\hat f^*(x) \approx 0$ then $\widehat{\mathrm{se}}(\hat f^*(x)) \approx 0$, so often $t^* = (\hat f^*(x) - \hat f(x))/\widehat{\mathrm{se}}(\hat f^*(x)) \approx \pm\infty$ and the interval may be far too wide.

11.4.2 Confidence bands

A confidence band controls at the confidence level the probability that the value $f(x)$ lies in the band at every position x, so it uses not the distribution of $\hat f(x) - f(x)$ but that of $\Delta = \sup_x |\hat f(x) - f(x)|$.[12] The distribution of Δ is hard to know exactly, and its asymptotic distribution is not normal either, so only fully nonparametric bootstrap bands are obtained.

The distribution of Δ is bootstrap-approximated from the bootstrap replicates $\hat f^*(x)$ of the kernel density estimate $\hat f(x)$ as follows. The function values are generally computed only on a grid $\mathcal{X} = \{x_1, \ldots, x_M\}$ finely dividing the interval containing the observations, so $\sup_x$ of Δ is approximated by $\max_{x\in\mathcal{X}}$. With the bootstrap replicates of Δ

$$\Delta^{*1} = \max_{x\in\mathcal{X}} |\hat f^{*1}(x) - \hat f(x)|, \ \ldots, \ \Delta^{*B} = \max_{x\in\mathcal{X}} |\hat f^{*B}(x) - \hat f(x)|$$

and their $1-\alpha$ quantile $Q_{\Delta^*}(1-\alpha)$, the bootstrap band is, for all x,

$$f_h(x) \in (\hat f(x) - Q_{\Delta^*}(1-\alpha),\ \hat f(x) + Q_{\Delta^*}(1-\alpha)).$$

[12] As multiple testing of finitely many hypotheses at once uses the distribution of the maximum of the test statistics to control the overall error rate, here the band is built from the distribution of the maximum of $|\hat f(x) - f(x)|$ over infinitely many $x \in \mathbb{R}$.

As $n \to \infty$, $h \to 0$, $nh \to \infty$, $B \to \infty$, this band is asymptotically exact:

$$P\big[f_h(x) \in (\hat{f}(x) - Q_{\Delta^*}(1-\alpha),\ \hat{f}(x) + Q_{\Delta^*}(1-\alpha)),\ \forall x\big] = 1 - \alpha + o(1).$$

Example 11.9

A band using only bootstrapped absolute deviations. In the code below, `abs.dev.boot` is the $M \times B$ matrix computed for the absolute-deviation bootstrap intervals in Example 11.8.

```
Delta = apply(abs.dev.boot,2,max)
q.Delta = quantile(Delta, 1-0.05)
conf.band.lower = f.hat - q.Delta
conf.band.upper = f.hat + q.Delta
```

The band is generally much wider than the pointwise intervals. Figure 11.12 of the book shows the bootstrap band together with the pointwise absolute-deviation bootstrap intervals.

Sometimes the lower limit of a band or pointwise interval is negative. A density cannot be negative, so replacing the lower limit $l(x)$ of a band (or interval) by $\max\{0, l(x)\}$ does not change the coverage. E.g. the 95% band of Figure 11.12 can be adjusted as in Figure 11.13 of the book.

Remark

When the interval at each x of the band is very wide, even if $\hat{f}(x)$ is near 0 and the pointwise interval is narrow, a very wide band results. To prevent this, one may take the maximum not of $|\hat{f}^{*b}(x) - \hat{f}(x)|$ but of the studentized $|\hat{f}^{*b}(x) - \hat{f}(x)|/\widehat{\text{se}}(\hat{f}^{*b}(x))$; but the variance of the bootstrap replicates is very large (consider $\hat{f}(x) \approx 0$), so this is not recommended. In practice, pointwise intervals are used more often than "simultaneous" bands covering all $x \in \mathcal{X}$. Some authors call the pointwise intervals at all $x \in \mathcal{X}$ a "confidence band"; take care.